

BIG DATA Y MACHINE LEARNING



A gentle introduction to R: From raw data to results

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A gentle introduction to R: From raw data to results

Data analysis is a workflow: we import data, learn what it contains, make **explicit** cleaning decisions, summarize it, communicate the findings, and export our work. In this session, we will follow that complete workflow with a sample from Colombia's *Gran Encuesta Integrada de Hogares* (GEIH)¹. Our practical question is:

What do weekly hours and monthly labor income look like among employed adults in Bogotá, and how do they differ by sex and employment formality?

Learning objectives

By the end of the session, you should be able to:

1. Import a dataset and inspect it.
2. Select, rename, recode and filter variables.
3. Produce general and grouped descriptive statistics.
4. Plot `ggplot2` figures.
5. Export tables and figures using reproducible project paths.

Additionally, we'll save everything in a GitHub repository, so you will learn:

1. what makes a repository reproducible.
2. how to organize the analysis in a self-contained R project.

3. how to track code changes using Git.
4. how to collaborate with a team using GitHub.
5. to document the repository using a [README](#).

Acknowledgement: Most of the materials used in this complementary session were contributed by previous TAs, especially [Eduard Martínez](#), [Lucas Gómez](#), Andrés Rengifo, Valentina Laverde, and Gustavo Castillo. We are truly grateful for their help.

Initial configuration

To load the required packages, we are using the `pacman` package, that simplifies the package management task.

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```
1 # Install pacman if necessary, then load the packages used in this session.
2 if (!require(pacman)) install.packages("pacman")
3 library(pacman)
4
5 p_load(
6   rio,          # Import data from a URL.
7   tidyverse,    # Manipulate data and create figures.
8   here,         # Build reproducible project paths.
9   skimr,        # Inspect the contents of a database.
10  gtsummary,     # Create descriptive tables.
11  gt,            # Format and export tables.
12  conflicted,    # Resolve conflicts between functions with the same name.
13  webshot2,      # Export gt tables as PNG files.
14  sessioninfo    # Show the session information.
15 )
16
17 conflict_prefer(name = "filter", winner = "dplyr")
18
19 options(scipen = 999) # Turn off scientific notation
```

Importing data

We are going to import a portion of the GEIH data that will be used in the first problem set. To do that, we will download a RDS file².

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```
1 db_geih <- rio::import("https://raw.githubusercontent.com/ignaciomsarmiento/datasets/main/GE
2   as_tibble()
```

```
3
4 head(db_geih)
```

Var.1	directorio	secuencia_p	orden	clase	dominio	m...	estrato1	sex
<int>	<int>	<int>	<int>	<int>	<chr>	<int>	<int>	<int>
1	4514331	1	2	1	BOGOTA	1	2	0
2	4514331	1	1	1	BOGOTA	1	2	1
3	4514332	1	4	1	BOGOTA	1	2	1
4	4514332	1	3	1	BOGOTA	1	2	1
5	4514332	1	1	1	BOGOTA	1	2	0
6	4514332	1	2	1	BOGOTA	1	2	1

6 rows | 1-9 of 178 columns

The same idea applies to other common formats:

Format	Typical function	Package
CSV	<code>read_csv("file.csv")</code>	<code>readr</code> / <code>tidyverse</code>
Excel	<code>read_excel("file.xlsx")</code>	<code>readxl</code>
Stata	<code>read_dta("file.dta")</code>	<code>haven</code>
SPSS	<code>read_sav("file.sav")</code>	<code>haven</code>
RDS	<code>readRDS("file.rds")</code>	Base R

The sample contains people interviewed in Bogotá. The combination of household, sequence, and person-order identifiers distinguishes individuals. The [data dictionary](#) documents the survey variables.

Inspecting the dataset

Never start cleaning before learning what a database contains. `glimpse()` emphasizes column names, types, and shows you a quick look at the data. `summary()` describes the distribution of selected variables. For now, let's focus on the variables that are important for our research question.

▼ Show/hide code

```
1 db_geih |>
2   select(age, sex, ocu, formal, totalHoursWorked, y_total_m) |>
3   glimpse()
```

```

Rows: 3,217
Columns: 6
$ age      <int> 29, 36, 4, 7, 32, 35, 18, 49, 51, 23, 10, 18, 12, 45,...
$ sex      <int> 0, 1, 1, 1, 0, 1, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 0, 1,...
$ ocu      <int> 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0,...
$ formal   <int> NA, 1, NA, NA, NA, NA, NA, NA, 1, NA, NA, NA, NA, 1, ...
$ totalHoursWorked <int> NA, 45, NA, NA, NA, NA, NA, NA, 48, NA, NA, NA, NA, 5...
$ y_total_m <dbl> NA, 1620833, NA, NA, NA, NA, NA, NA, 1378973, NA, NA,...

```

▼ Show/hide code

```
1 summary(db_geih |> select(age, sex, ocu, formal, totalHoursWorked, y_total_m))
```

```

      age      sex      ocu      formal
Min.   : 0.00  Min.   :0.0000  Min.   :0.0000  Min.   :0.0000
1st Qu.:18.00  1st Qu.:0.0000  1st Qu.:0.0000  1st Qu.:0.0000
Median :32.00  Median :0.0000  Median :0.0000  Median :1.0000
Mean   :34.52  Mean   :0.4756  Mean   :0.4865  Mean   :0.5834
3rd Qu.:51.00  3rd Qu.:1.0000  3rd Qu.:1.0000  3rd Qu.:1.0000
Max.   :95.00  Max.   :1.0000  Max.   :1.0000  Max.   :1.0000
                        NA's   :1652

totalHoursWorked  y_total_m
Min.   :  2.00  Min.   :  40000
1st Qu.: 40.00  1st Qu.: 769913
Median : 48.00  Median : 937512
Mean   : 47.16  Mean   :1437443
3rd Qu.: 50.00  3rd Qu.:1499042
Max.   :130.00  Max.   :40000000
NA's   :1652    NA's   :1863

```

For a compact but richer diagnostic, `skim()` combines type, missingness, and distribution information.

▼ Show/hide code

```

1 db_geih |>
2   select(age, sex, ocu, formal, totalHoursWorked, y_total_m) |>
3   skimr::skim()

```

Data summary

Name	select(db_geih, age, sex,...
Number of rows	3217
Number of columns	6
Column type frequency:	
numeric	6

Group variables

None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75
age	0	1.00	34.52	21.03	0	18.0	32.0	51
sex	0	1.00	0.48	0.50	0	0.0	0.0	1
ocu	0	1.00	0.49	0.50	0	0.0	0.0	1
formal	1652	0.49	0.58	0.49	0	0.0	1.0	1
totalHoursWorked	1652	0.49	47.16	14.63	2	40.0	48.0	50
y_total_m	1863	0.42	1437443.20	1990665.19	40000	769912.8	937511.8	1499042

Note that the missing values in hours, formality, and labour income are not necessarily mistakes. These questions do not apply to children or people who are not employed. This is **structural missingness** that we must fix by defining our analysis unit before deciding which NA values to remove.

Cleaning the dataset

Our unit of analysis will be an employed adult. Cleaning is a sequence of explicit decisions³ made over the unit of analysis.

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```

1  # We can rename several columns by creating a vector where each entry corresponds
2  # to the name of a column and its index corresponds to the new name we want to assign.
3  text_vars <- c('num_age' = 'age', 'bin_male' = 'sex', 'bin_employed' = 'ocu',
4                'bin_formal' = 'formal', 'num_hoursWorked' = 'totalHoursWorked',
5                'num_labIncome' = 'y_total_m')
6  db_geih <- db_geih |>
7    select(all_of(text_vars)) |>
8    filter(num_age >= 18,
9           bin_employed == 1) |>
10 # The logical variable `bin_employed` and the age are used to define the sample
11 # and then removed.
12 select(-c('num_age', 'bin_employed')) |>
13 mutate(bin_male = factor(bin_male, levels = 0:1, labels = c('Female', 'Male')),
14        bin_formal = factor(bin_formal, levels = 0:1, labels = c('Informal', 'Formal')))
15

```

```
16 # Given our data handling, let's check again the distribution.
17 db_geih |> skimr::skim()
```

Data summary

Name	db_geih
Number of rows	1551
Number of columns	4
Column type frequency:	
factor	2
numeric	2
Group variables	
None	

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
bin_male	0	1	FALSE	2	Mal: 826, Fem: 725
bin_formal	0	1	FALSE	2	For: 913, Inf: 638

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75
num_hoursWorked	0	1.00	47.27	14.5	2	40.0	48.0	50
num_labIncome	209	0.87	1446471.97	1997059.6	40000	781060.5	941845.2	1500000

Even though we’re filtering by employed adults, approximately 13.5% of the observations lack labor income. Your mission, if you choose to accept it, is to determine why that happened. We should not just drop **NA** observations without arguing why it’s a proper choice. It’s important that you understand the population and the dataset before making methodological decisions about it. Otherwise, it is not possible to trust the statistics you present. For now, we can just drop **NA** observations (or use the argument `na.rm = TRUE` when needed). However, beware that this solution only avoids the computational matter. It estimates values among reporters, but doesn’t solve the selection issue.

▼ Show/hide code

```
1 db_geih <- db_geih |> drop_na()
```

Descriptive statistics

Descriptive statistics reduce many observations to a small set of interpretable quantities. Let's remember our research question. What can we say about weekly hours and monthly labor income across employed adults in Bogotá? How do they differ by sex and employment formality?

General summaries

▼ Show/hide code

```
1 tab_example <- db_geih |>
2   summarise(mean_num_hours = mean(num_hoursWorked),
3             median_num_hours = median(num_hoursWorked),
4             mean_num_income = mean(num_labIncome),
5             median_num_income = median(num_labIncome))
```

`across()` to apply several summary functions to different variables

Grouped summaries with `summarise()`

▼ Show/hide code

```
1 tab_example <- db_geih |>
2   group_by(bin_male, bin_formal) |>
3   summarise(num_obs = n(),
4             mean_num_income = mean(num_labIncome),
5             sd_num_income = sd(num_labIncome),
6             p25_num_income = quantile(num_labIncome, 0.25),
7             p75_num_income = quantile(num_labIncome, 0.75),
8             .groups = "drop")
```

Given the previous result, we can create a table with the `gt` package. However, we'll see that there's a more efficient way to do this with the `gtsummary` package.

▼ Show/hide code

```
1 tab_example |>
2   gt() |>
3   cols_label(bin_male = "Sex",
4             bin_formal = "Type of employment",
5             num_obs = "N",
6             mean_num_income = "Mean income",
7             sd_num_income = "SD",
```

```

8       p25_num_income = "P25",
9       p75_num_income = "P75") |>
10    fmt_integer(columns = num_obs, sep_mark = ",") |>
11    fmt_number(columns = starts_with("num_") & -num_obs,
12              decimals = 0,
13              sep_mark = ",")

```

Sex	Type of employment	N	Mean income	SD	P25	P75
Female	Informal	261	758666.9	1319898.2	350000.0	806000
Female	Formal	377	1809078.8	2067440.1	882333.4	1866667
Male	Informal	273	991179.2	785850.8	600000.0	1200000
Male	Formal	431	1834196.5	2563492.9	911333.4	1701833

A presentation-ready table

For the following tables, we won't provide the code. Part of the work in this course is for you to program your own code! Code that you understand and that helps you generate descriptive statistics tables, regression model results tables, etc. However, let's check what we usually expect from the descriptive statistics.

Table 1. Descriptive statistics by formality.

Variable	Total N = 1,342	Type of employment		p-value ²
		Informal N = 534 ¹	Formal N = 808 ¹	
Sex				0.43
Female	638 (48%)	261 (49%)	377 (47%)	
Male	704 (52%)	273 (51%)	431 (53%)	
Weekly hours worked	47 (14)	45 (18)	49 (11)	<0.001

¹ For continuous variables, mean (sd). For categorical, frequency (percentage).

² H0: Equal means.

Source: Own calculations using the GEIH sample.

Variable	Total N = 1,342	Type of employment		p-value ²
		Informal N = 534 ¹	Formal N = 808 ¹	
Monthly labour income (COP)	1,446,472 (1,997,060)	877,536 (1,085,591)	1,822,477 (2,343,757)	<0.001

¹ For continuous variables, mean (sd). For categorical, frequency (percentage).

² H0: Equal means.

Source: Own calculations using the GEIH sample.

Interpretation

How can we interpret the p -values? Can you establish if this difference is economically important? Is this estimate an association or a causal relationship?

```
Table 1 Column Name  Header
label                "***Characteristic**"
n                    "***N**"
stat_0               "***Overall**  \nN = 638"
stat_1               "***Informal**  \nN = 261"
stat_2               "***Formal**  \nN = 377"
p.value              "***p-value**"
Table 2 Column Name  Header
label                "***Characteristic**"
n                    "***N**"
stat_0               "***Overall**  \nN = 704"
stat_1               "***Informal**  \nN = 273"
stat_2               "***Formal**  \nN = 431"
p.value              "***p-value**"
```

Table 2. Average labor income across different groups.

Average labor income	N	Overall ¹	Type of employment		p-value ²
			Informal ¹	Formal ¹	
Sex					
Female	638	1,379,365 (1,871,149)	758,667 (1,319,898)	1,809,079 (2,067,440)	<0.001

¹ For continuous variables, mean (sd).

² H0: Equal means.

Average labor income	N	Overall ¹	Type of employment		p-value ²
			Informal ¹	Formal ¹	
Male	704	1,507,288 (2,104,151)	991,179 (785,851)	1,834,197 (2,563,493)	<0.001

¹ For continuous variables, mean (sd).

² H0: Equal means.

gt and gtsummary

Visualizing data with ggplot2

We can make graphs in several ways, but `ggplot2` is one of the most elegant and most versatile. The “gg” in *ggplot* stand for Grammar of Graphics, and it’s coherent system that allows you to compose graphs by adding new layers.

1. `ggplot()` creates a blank canvas.
2. A `geom_*()` function determines how observations are drawn.
3. Scales control how values are displayed.
4. `labs()` and themes make the result self-contained.

Distribution of labor income

▼ Show/hide code

```

1  # Scott's rule.
2  num_binwidth <- 3.5 * sd(db_geih$num_labIncome) / length(db_geih$num_labIncome)^(1/3)
3  num_binwidth <- round(x = num_binwidth, digits = 2)
4
5  fig_income_distribution <- db_geih |>
6    ggplot(mapping = aes(x = num_labIncome)) +
7    geom_histogram(mapping = aes(y = (after_stat(count))/sum(after_stat(count))),
8                    color = "#FFFFFF", fill = "#3a5e8cFF", show.legend = FALSE, na.rm = TRUE,
9                    binwidth = num_binwidth, alpha = 0.8, closed = 'left') +
10   coord_cartesian(xlim = c(0, quantile(db_geih$num_labIncome, probs = 0.99))) +
11   scale_x_continuous(
12     labels = scales::label_number(scale = 0.000001),
13     expand = expansion(mult = c(0, 0.05))
14   ) +
15   scale_y_continuous(
16     expand = expansion(mult = c(0, 0.05)),

```

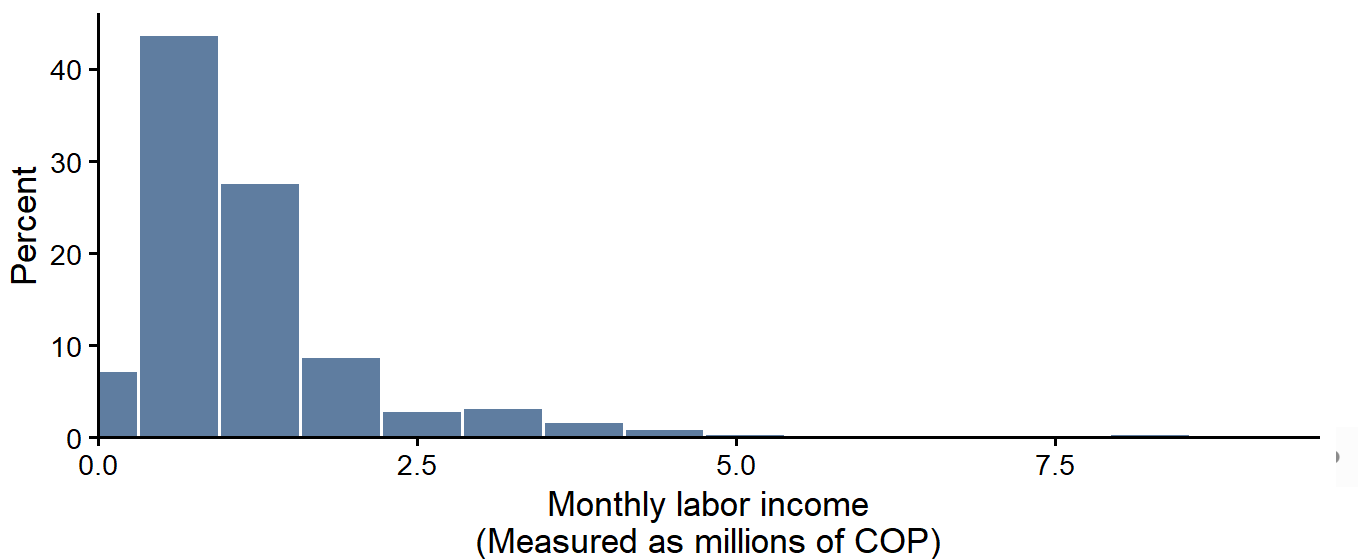
```

17   limits = c(0, NA),
18   labels = scales::label_number(scale = 100)
19 ) +
20 labs(
21   title = "Distribution of monthly labor income among employed adults",
22   subtitle = "",
23   x = "Monthly labor income\n(Measured as millions of COP)",
24   y = "Percent",
25   caption = paste0(
26     "Source. Own calculations using the GEIH sample.\n",
27     "Note. Binwidth calculated through Scott's rule. The x-axis is zoomed at the 99th per
28   )
29 )
30
31 fig_income_distribution

```

Distribution of monthly labor income among employed adults.

Distribution of monthly labor income among employed adults



Source. Own calculations using the GEIH sample.

Note. Binwidth calculated through Scott's rule. The x-axis is zoomed at the 99th percentile.

Income by formality and sex

▼ Show/hide code

```

1 fig_income_by_sex_and_formality <- db_geih |>
2   ggplot(mapping = aes(x = bin_formal, y = num_labIncome, fill = factor(1))) +
3   geom_boxplot(width = 0.65, alpha = 0.65, outlier.alpha = 0.25,
4               show.legend = FALSE, na.rm = TRUE) +

```

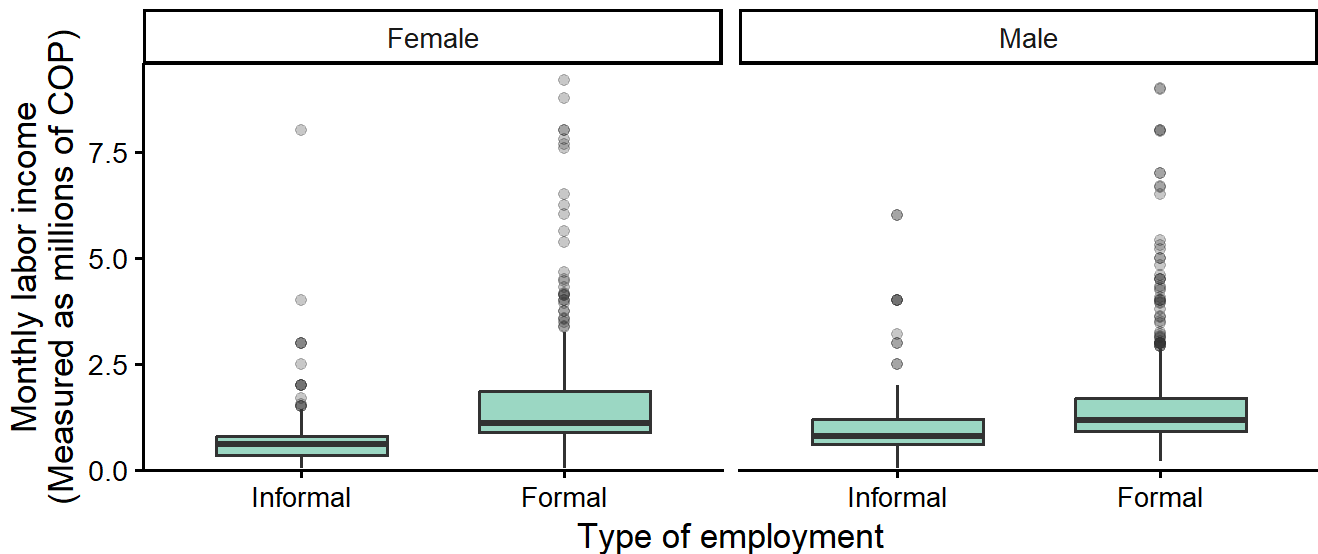
```

5   facet_wrap(vars(bin_male)) +
6   coord_cartesian(ylim = c(0, quantile(db_geih$num_labIncome, probs = 0.99))) +
7   scale_fill_brewer(palette = "Set2") +
8   scale_y_continuous(
9     labels = scales::label_number(scale = 0.000001),
10    expand = expansion(mult = c(0, 0.05))
11  ) +
12  labs(
13    title = "Labor income across sex and type of employment",
14    subtitle = "",
15    x = "Type of employment",
16    y = "Monthly labor income\n(Measured as millions of COP)",
17    caption = paste0(
18      "Source. Own calculations using the GEIH sample.\n",
19      "Note. The y-axis is zoomed at the 99th percentile."
20    )
21  )
22
23  fig_income_by_sex_and_formality

```

Monthly labor income by employment formality and sex.

Labor income across sex and type of employment



Source. Own calculations using the GEIH sample.
 Note. The y-axis is zoomed at the 99th percentile.

Exporting results

Saving a named object makes export reproducible: we export the exact table or figure that we inspected above. We shouldn't edit figures manually. For now, we'll export everything as a [PNG](#), but

you can export plots as **PDF** (for higher resolution) and tables as **TEX** (to include it in your LaTeX document).

▼ Show/hide code

```
1 ggsave(  
2   filename = here("output", "figures", "fig_income_distribution.png"),  
3   plot = fig_income_distribution,  
4   width = 7,  
5   height = 4.5,  
6   dpi = 300,  
7   bg = "transparent"  
8 )  
9  
10 ggsave(  
11   filename = here("output", "figures", "fig_income_by_sex_and_formality.png"),  
12   plot = fig_income_by_sex_and_formality,  
13   width = 7,  
14   height = 4.5,  
15   dpi = 300,  
16   bg = "transparent"  
17 )  
18  
19 gtsave(  
20   data = tab_descriptive,  
21   filename = here::here("output", "tables", "tab_descriptive.png"),  
22   zoom = 2  
23 )
```

Conclusions

Before finishing, ask yourself:

- Do I understand the dataset? What the unit of analysis is?
- Can another person understand my work without further information?
- Does each table or figure answer a clear question?
- Can the repository recreate every exported result?

References and further practice

- Wickham, H., Çetinkaya-Rundel, M., and Grolemund, G. (2023). [R for Data Science \(2e\)](#) .
- [ggplot2 documentation](#) .
- [GEIH sample data dictionary](#) .

Session information

`session_info()` reports the R version, platform, locale, and loaded package versions used to execute this notebook. This information can help diagnose differences across computers.

▼ Show/hide code

```
1 sessioninfo::session_info()
```

— Session info —

setting	value
version	R version 4.5.1 (2025-06-13 ucrt)
os	Windows 10 x64 (build 19045)
system	x86_64, mingw32
ui	RTerm
language	(EN)
collate	Spanish_Colombia.utf8
ctype	Spanish_Colombia.utf8
tz	America/Bogota
date	2026-08-07
pandoc	3.4 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
quarto	1.6.42 @ C:\PROGRA~1\RStudio\RESOUR~1\app\bin\quarto\bin\quarto.exe

— Packages —

package	* version	date (UTC)	lib	source
backports	1.5.1	2026-04-03	[1]	CRAN (R 4.5.3)
base64enc	0.1-6	2026-02-02	[1]	CRAN (R 4.5.2)
broom	1.0.13	2026-05-14	[1]	CRAN (R 4.5.3)
cachem	1.1.0	2024-05-16	[1]	CRAN (R 4.5.1)
cards	0.8.1	2026-07-06	[1]	CRAN (R 4.5.3)
cardx	0.3.4	2026-07-06	[1]	CRAN (R 4.5.3)
chromote	0.5.1	2025-04-24	[1]	CRAN (R 4.5.3)
cli	3.6.6	2026-04-09	[1]	CRAN (R 4.5.3)
commonmark	2.0.0	2025-07-07	[1]	CRAN (R 4.5.1)
conflicted	* 1.2.0	2023-02-01	[1]	CRAN (R 4.5.1)
curl	7.1.0	2026-04-22	[1]	CRAN (R 4.5.3)
digest	0.6.39	2025-11-19	[1]	CRAN (R 4.5.1)
dplyr	* 1.2.1	2026-04-03	[1]	CRAN (R 4.5.3)
evaluate	1.0.5	2025-08-27	[1]	CRAN (R 4.5.1)
farver	2.1.2	2024-05-13	[1]	CRAN (R 4.5.1)
fastmap	1.2.0	2024-05-15	[1]	CRAN (R 4.5.1)
forcats	* 1.0.1	2025-09-25	[1]	CRAN (R 4.5.1)
fs	2.1.0	2026-04-18	[1]	CRAN (R 4.5.3)
generics	0.1.4	2025-05-09	[1]	CRAN (R 4.5.1)
ggplot2	* 4.0.3	2026-04-22	[1]	CRAN (R 4.5.3)
glue	1.8.1	2026-04-17	[1]	CRAN (R 4.5.3)
gt	* 1.3.0	2026-01-22	[1]	CRAN (R 4.5.3)
gtable	0.3.6	2024-10-25	[1]	CRAN (R 4.5.1)

gtsummary	* 2.5.1	2026-05-30	[1]	CRAN (R 4.5.3)
here	* 1.0.2	2025-09-15	[1]	CRAN (R 4.5.1)
hms	1.1.4	2025-10-17	[1]	CRAN (R 4.5.1)
htmltools	0.5.9	2025-12-04	[1]	CRAN (R 4.5.2)
htmlwidgets	1.6.4	2023-12-06	[1]	CRAN (R 4.5.1)
jsonlite	2.0.0	2025-03-27	[1]	CRAN (R 4.5.1)
knitr	1.51	2025-12-20	[1]	CRAN (R 4.5.2)
labeling	0.4.3	2023-08-29	[1]	CRAN (R 4.5.0)
later	1.4.8	2026-03-05	[1]	CRAN (R 4.5.3)
lifecycle	1.0.5	2026-01-08	[1]	CRAN (R 4.5.2)
litedown	0.10	2026-07-11	[1]	CRAN (R 4.5.3)
lubridate	* 1.9.5	2026-02-04	[1]	CRAN (R 4.5.3)
magrittr	2.0.5	2026-04-04	[1]	CRAN (R 4.5.3)
markdown	2.0	2025-03-23	[1]	CRAN (R 4.5.1)
memoise	2.0.1	2021-11-26	[1]	CRAN (R 4.5.1)
otel	0.2.0	2025-08-29	[1]	CRAN (R 4.5.1)
pacman	* 0.5.1	2019-03-11	[1]	CRAN (R 4.5.1)
pillar	1.11.1	2025-09-17	[1]	CRAN (R 4.5.1)
pkgconfig	2.0.3	2019-09-22	[1]	CRAN (R 4.5.1)
processx	3.9.0	2026-04-22	[1]	CRAN (R 4.5.3)
promises	1.5.0	2025-11-01	[1]	CRAN (R 4.5.2)
purrr	* 1.2.2	2026-04-10	[1]	CRAN (R 4.5.3)
R.methodsS3	1.8.2	2022-06-13	[1]	CRAN (R 4.5.0)
R.oo	1.27.1	2025-05-02	[1]	CRAN (R 4.5.0)
R.utils	2.13.0	2025-02-24	[1]	CRAN (R 4.5.1)
R6	2.6.1	2025-02-15	[1]	CRAN (R 4.5.1)
ragg	1.5.2	2026-03-23	[1]	CRAN (R 4.5.3)
RColorBrewer	1.1-3	2022-04-03	[1]	CRAN (R 4.5.0)
Rcpp	1.1.2	2026-07-05	[1]	CRAN (R 4.5.3)
readr	* 2.2.0	2026-02-19	[1]	CRAN (R 4.5.3)
repr	1.1.7	2024-03-22	[1]	CRAN (R 4.5.1)
rio	* 1.3.0	2026-04-02	[1]	CRAN (R 4.5.3)
rlang	1.3.0	2026-07-05	[1]	CRAN (R 4.5.3)
rmarkdown	2.31	2026-03-26	[1]	CRAN (R 4.5.3)
rprojroot	2.1.1	2025-08-26	[1]	CRAN (R 4.5.1)
rstudioapi	0.19.0	2026-06-11	[1]	CRAN (R 4.5.3)
S7	0.2.2	2026-04-22	[1]	CRAN (R 4.5.3)
sass	0.4.10	2025-04-11	[1]	CRAN (R 4.5.1)
scales	1.4.0	2025-04-24	[1]	CRAN (R 4.5.1)
sessioninfo	* 1.2.4	2026-06-04	[1]	CRAN (R 4.5.3)
skimr	* 2.2.2	2026-01-10	[1]	CRAN (R 4.5.2)
stringi	1.8.7	2025-03-27	[1]	CRAN (R 4.5.0)
stringr	* 1.6.0	2025-11-04	[1]	CRAN (R 4.5.2)
systemfonts	1.3.2	2026-03-05	[1]	CRAN (R 4.5.3)
textshaping	1.0.5	2026-03-06	[1]	CRAN (R 4.5.3)
tibble	* 3.3.1	2026-01-11	[1]	CRAN (R 4.5.2)
tidyr	* 1.3.2	2025-12-19	[1]	CRAN (R 4.5.2)
tidyselect	1.2.1	2024-03-11	[1]	CRAN (R 4.5.1)
tidyverse	* 2.0.0	2023-02-22	[1]	CRAN (R 4.5.1)
timechange	0.4.0	2026-01-29	[1]	CRAN (R 4.5.3)
tzdb	0.5.0	2025-03-15	[1]	CRAN (R 4.5.1)

vctrs	0.7.3	2026-04-11	[1]	CRAN (R 4.5.3)
webshot2	* 0.1.2	2025-04-23	[1]	CRAN (R 4.5.3)
websocket	1.4.4	2025-04-10	[1]	CRAN (R 4.5.3)
withr	3.0.3	2026-06-19	[1]	CRAN (R 4.5.3)
xfun	0.60	2026-07-09	[1]	CRAN (R 4.5.3)
xml2	1.6.0	2026-06-22	[1]	CRAN (R 4.5.3)
yaml	2.3.12	2025-12-10	[1]	CRAN (R 4.5.2)

[1] C:/Users/judro/AppData/Local/R/win-library/4.5

[2] C:/Program Files/R/R-4.5.1/library

* — Packages attached to the search path.

Footnotes

1. If you have comments or suggestions, please send an email. They are welcome and will be considered in your final grade. [↩](#)
2. An RDS file stores one R object. [↩](#)
3. If you already use agentic AI, please, don't let the agent take the methodological decisions for you. [↩](#)